

Inverse Frontiers for the Fabius–Rvachev System

Exact inverse-dyadic algebraic germs, Catalan–Gaussian-binomial acceleration, and periodic Lambert–Bell quantile asymptotics

Frontier research report prepared from the complete documentation corpus

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Corpus snapshot: 57 LaTeX sources under `Analysis/FabiusFunction/docs`, including frozen archive variants

Abstract

The documentation tree under `Analysis/FabiusFunction/docs` contains a broad, mutually cross-checked theory of the Thue–Morse sequence, the Fabius distribution function F , its totalized inverse $G = F^{-1}$, Rvachev’s compactly supported up-function, exact dyadic arithmetic, q -binomial formulae, infinite sinc products, and lower-Lambert endpoint asymptotics. This report audits every one of the 57 LaTeX sources in that tree and develops an inverse calculus that is not present in those documents.

There are two principal advances. At finite scale, the ordinary n -factor Thue–Morse spline P_n is shown to have, near every inverse-dyadic anchor $x_r = 2^{-r}$, an *exact* degree- r local polynomial obtained by applying a finite reciprocal-sinc operator to the terminating Taylor jet of F . Consequently the finite-prefix quantile $G_n(F(2^{-r}))$ is an algebraic germ in $Q = 4^{-n}$. At $r = 2$ the germ is the explicit square root

$$G_n(5/72) = \frac{1}{4} + \frac{\sqrt{1 + (64/9)4^{-n}} - 1}{8} \quad (n \geq 3),$$

whose full error series has Catalan coefficients. Lagrange interpolation on the geometric nodes $1, q, \dots, q^{s-1}$, $q = 1/4$, then yields Gaussian q -binomial Richardson filters; at the quarter anchor their leading constants are closed Catalan expressions. A general Bell-polynomial inverse-transfer theorem gives the first two correction functions at every interior quantile.

At the endpoint, the corpus gives the leading equivalent for $G(y)$ but only remarks that the nonconstant Gamma–zeta periodic fluctuation must reappear at the next order. We carry out that missing inversion. If $T = \log(1/y)$, $\rho = \sqrt{2T/\log 2}$, and $\theta = \rho + 1/\log 2 + 1/2$, then

$$\frac{G(y)}{G_0(y)} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right), \quad G_0(y) = \frac{\sqrt{T}}{e\sqrt{\log 2}} e^{-\sqrt{2T\log 2}}.$$

We give the second correction explicitly, an all-orders coefficient recursion in the differential algebra generated by the periodic saddle jets, a nondegenerate periodic limit-set theorem proving that no constant second term can exist, and an asymptotic formula for the quantile elasticity $yG'(y)/G(y)$.

All new statements are proved from results already established in the corpus. Exact rational, symbolic, and high-precision checks accompany the proofs. “New” here means new relative to the inspected repository corpus; no unconditional claim of global publication priority is made.

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1 Executive summary, status, and nonduplication boundary

1.1 What was inspected

The audit covered every file with extension `.tex` below `Analysis/FabiusFunction/docs`, recursively, using the living repository expositions as the primary mathematical interface [14, 15, 16, 17]. The current tree contains 25 living or embedded-paper sources and 32 frozen historical variants, for a total of 57. The living set consists of five primary expository documents, one active Thue–Morse formula atlas, twelve Fourier-decay drafts or audits, and seven primary-paper transcriptions. The archive contains early notes, superseded expositions, q -calculus supplements, inverse and endpoint studies, convergence reports, and three historical Atlas variants. The complete path ledger appears in [appendix A](#).

The mathematical audit was thematic rather than merely lexical. In particular, the search covered occurrences and nearby arguments involving

$$F^{-1}, \quad \text{quantiles}, \quad \text{terminating dyadic jets}, \quad \frac{1}{\prod \text{sinc}}, \quad \text{Bell polynomials},$$

$$\text{Lagrange interpolation}, \quad \text{Richardson extrapolation}$$

as well as the exact Lambert phase and the periodic Gamma–zeta term. The primary “Missing Parts” document proves the global inverse package and the leading endpoint equivalent, then states explicitly that periodicity returns in the next relative correction; it does not compute that correction. The finite-prefix reports develop reciprocal-sinc and q -Richardson expansions for F or up, but do not invert them and do not identify the exact algebraic quantile germs below.

1.2 Contribution map

Topic	Result in this report	Status
Interior inverse transfer	Uniform all-orders expansion of $G_n = P_n^{-1}$, with a Bell-polynomial recursion and explicit coefficients through 4^{-2n} .	New to corpus
Inverse-dyadic anchors	Exact local polynomial for P_n near 2^{-r} and exact algebraic germ for $G_n(F(2^{-r}))$.	New to corpus
Quarter anchor	Closed square-root quantile, convergent Catalan series, and exact finite-depth identity for all $n \geq 3$.	New to corpus
q -Richardson	Gaussian q -binomial filters applied to inverse approximants, with an exact moment identity for all residual powers.	New application
Endpoint inverse	First and second oscillatory corrections, all-orders Lambert–Bell reversion, and explicit periodic phase.	New to corpus
Limit set	Complete subsequential limit interval for the normalized second inverse term; impossibility of a constant correction.	New to corpus
Inverse derivatives	Two-term asymptotic for $yG'(y)/G(y)$ including Ψ' .	New to corpus

Table 1: The principal results and their corpus-relative status.

1.3 What is deliberately not repeated

The report does not re-prove the repository’s exact rational evaluator at arbitrary dyadic arguments, the many $q = 1/2$ Gaussian-binomial identities, the Thue–Morse formula atlas, the complete saddle analysis for F , the Fourier-decay orbit laws, or the positive Gaussian/Lobatto tail-quadrature hierarchies.

Only the minimum needed interfaces are recalled. Likewise, previous frontier work on arithmetic dyadic rays, cyclotomic orbit polynomials, finite Thue–Morse–sinc bridges, and spectral Richardson acceleration is treated as a nonduplication boundary rather than recycled as a contribution.

The core novelty is an *inverse* principle. A forward approximation with a geometric error germ does not merely give an error bound for its inverse. It carries a complete nonlinear coefficient algebra. At inverse-dyadic anchors the terminating derivatives and the exact Thue–Morse plateau collapse that nonlinear algebra to a finite polynomial, turning an asymptotic inverse problem into an exact algebraic one.

2 Established platform and notation

2.1 Fabius, Rvachev, and the random series

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$ and put

$$Y := \sum_{k=1}^{\infty} 2^{-k} U_k, \quad S_n := \sum_{k=1}^n 2^{-k} U_k. \quad (2.1)$$

The density of Y is Rvachev’s even up-function up, supported on $[-1, 1]$ [2, 3, 14]. The bounded Fabius function, originating in Fabius’s probabilistic construction [1], satisfies

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (2.2)$$

and is a strictly increasing C^∞ bijection from $[0, 1]$ to itself. We write

$$G := F^{-1} : [0, 1] \longrightarrow [0, 1]. \quad (2.3)$$

The reflection laws are

$$F(1 - x) = 1 - F(x), \quad G(1 - y) = 1 - G(y). \quad (2.4)$$

Let p_n be the density of S_n and set

$$P_n(x) := p_n(x - 1), \quad 0 \leq x \leq 1. \quad (2.5)$$

On every compact subinterval of $(0, 1)$, P_n is strictly increasing for all sufficiently large n . Its increasing inverse branch will be denoted

$$G_n := P_n^{-1}. \quad (2.6)$$

The omitted random tail is a scaled copy of Y :

$$Y - S_n \stackrel{d}{=} 2^{-n} Y. \quad (2.7)$$

2.2 The finite Thue–Morse spline

Put $\varepsilon_j = (-1)^{s_2(j)}$, the signed Thue–Morse sequence [6]. The unequal-box inclusion–exclusion formula for the corresponding box spline [11] gives

$$P_n(x) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \varepsilon_j (x - 2^{-n} - j2^{1-n})_+^{n-1}. \quad (2.8)$$

Thus every $G_n(y)$ is the inverse of an explicitly known piecewise polynomial. Formula (2.8) alone does not reveal which polynomial piece controls a prescribed quantile or why its coefficients should simplify. The local theorem in [section 4](#) answers both questions at the inverse-dyadic outputs $y = F(2^{-r})$.

2.3 Fourier product and reciprocal-sinc coefficients

With the Fourier convention $\widehat{f}(\xi) = \int f(x)e^{-ix\xi} dx$,

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{k=1}^{\infty} \text{sinc}(2^{-k}\xi), \quad \widehat{p}_n(\xi) = \prod_{k=1}^n \text{sinc}(2^{-k}\xi), \quad \text{sinc } t := \frac{\sin t}{t}, \quad \text{sinc } 0 := 1. \quad (2.9)$$

Near the origin define rational coefficients b_j by

$$\frac{1}{\Phi(\xi)} = \sum_{j=0}^{\infty} b_j \xi^{2j}, \quad b_0 = 1. \quad (2.10)$$

Euler's product gives

$$\log \frac{1}{\Phi(\xi)} = \sum_{k=1}^{\infty} \gamma_k \xi^{2k}, \quad \gamma_k = \frac{\zeta(2k)}{k\pi^{2k}(4^k - 1)}. \quad (2.11)$$

Equivalently, in Bernoulli form,

$$\gamma_k = \frac{(-1)^{k+1} 2^{2k-1} B_{2k}}{k(2k)!(4^k - 1)}. \quad (2.12)$$

The coefficient array is generated by complete Bell polynomials [10]:

$$b_j = \frac{1}{j!} \mathcal{B}_j(1!\gamma_1, 2!\gamma_2, \dots, j!\gamma_j), \quad (2.13)$$

or by the recurrence

$$b_j = \frac{1}{j} \sum_{k=1}^j k \gamma_k b_{j-k}. \quad (2.14)$$

The first values are

$$b_1 = \frac{1}{18}, \quad b_2 = \frac{31}{16200}, \quad b_3 = \frac{2347}{42865200}, \quad b_4 = \frac{1904369}{1311675120000}. \quad (2.15)$$

The corpus proves the uniform derivative expansion

$$P_n^{(m)}(x) = \sum_{j=0}^{J-1} (-1)^j b_j 4^{-jn} F^{(m+2j)}(x) + \mathcal{O}_{m,J}(4^{-Jn}) \quad (2.16)$$

uniformly on $[0, 1]$ for every fixed m, J . We shall invert this expansion.

2.4 Terminating jets at inverse powers of two

Let

$$x_r := 2^{-r}, \quad y_r := F(x_r). \quad (2.17)$$

The global dilation identity implies

$$F^{(k)}(x_r) = 2^{\binom{k+1}{2}} F(2^{k-r}), \quad 0 \leq k \leq r, \quad F^{(k)}(x_r) = 0 \quad (k > r). \quad (2.18)$$

Write $d_0 = 1$ and

$$d_r = r! 2^{\binom{r}{2}} F(2^{-r}). \quad (2.19)$$

Then the nonconstant part of the terminating Taylor jet is

$$J_r(z) := \sum_{k=1}^r \alpha_{r,k} z^k, \quad \alpha_{r,k} = 2^{\binom{k+1}{2} - \binom{r-k}{2}} \frac{d_{r-k}}{k!(r-k)!}. \quad (2.20)$$

Thus the Taylor series of F at x_r is the polynomial $y_r + J_r(z)$, although $F(x_r + z)$ is not equal to that polynomial on any neighborhood. The discrepancy is flat at $z = 0$.

3 Bell-polynomial transfer from forward splines to quantiles

3.1 A general inverse perturbation theorem

The first new step is abstract. It is useful well beyond the Fabius function. Let $q = 1/4$ and abbreviate $Q_n = q^n$. From (2.16),

$$P_n(x) = F(x) + \sum_{j=1}^{J-1} a_j(x) Q_n^j + \mathcal{O}(Q_n^J), \quad a_j(x) = (-1)^j b_j F^{(2j)}(x). \quad (3.1)$$

Theorem 3.1 (Uniform inverse-transfer expansion). *Fix $0 < \eta < 1/2$ and an integer $J \geq 1$. There exist smooth functions $c_1, \dots, c_{J-1}$ on $[\eta, 1 - \eta]$ such that*

$$G_n(y) = G(y) + \sum_{j=1}^{J-1} c_j(y) Q_n^j + \mathcal{O}_{\eta,J}(Q_n^J) \quad (3.2)$$

uniformly for $y \in [\eta, 1 - \eta]$. Put $x = G(y)$ and $X_m = m!c_m(y)$. With $\mathcal{B}_{m,k}$ denoting the partial exponential Bell polynomial and $\mathcal{B}_{0,0} = 1$, the coefficients obey

$$c_m(y) = -\frac{1}{F'(x)} \left[\frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(X_1, \dots, X_{m-k+1}) + \sum_{j=1}^m \frac{1}{(m-j)!} \sum_{k=0}^{m-j} a_j^{(k)}(x) \mathcal{B}_{m-j,k}(X_1, \dots, X_{m-j-k+1}) \right]. \quad (3.3)$$

In the second line, the term with $m = j$ is interpreted as $a_m(x)$.

Proof. On the compact interval $K = G([\eta, 1 - \eta]) \subseteq (0, 1)$, F' has a positive minimum. The derivative version of (2.16) therefore implies $P'_n > 0$ on a fixed neighborhood of K for all large n , so the inverse branch G_n exists there.

Write $x = G(y)$ and

$$\delta_n(y) := G_n(y) - x = \sum_{m \geq 1} c_m(y) Q_n^m.$$

Substitute $x + \delta_n$ into (3.1) and use $P_n(x + \delta_n) = F(x)$. The coefficient of Q_n^m in $F(x + \delta_n) - F(x)$ is

$$F'(x)c_m + \frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(1!c_1, 2!c_2, \dots).$$

The coefficient of Q_n^{m-j} in $a_j(x + \delta_n)$ is the corresponding second Bell sum in (3.3). Solving the resulting triangular equation for c_m gives the formula. Taylor's theorem with a uniform remainder and the lower bound for F' convert the formal cancellation through order $J - 1$ into the uniform remainder in (3.2). $\square$

3.2 The first two quantile corrections

The first coefficient already has a simple geometric meaning: it is the logarithmic derivative of the density scale.

Corollary 3.2 (Explicit coefficients through 16^{-n}). *At $x = G(y)$,*

$$c_1(y) = \frac{1}{18} \frac{F''(x)}{F'(x)} \quad (3.4)$$

and

$$c_2(y) = -\frac{31}{16200} \frac{F^{(4)}(x)}{F'(x)} + \frac{1}{324} \frac{F''(x)F'''(x)}{F'(x)^2} - \frac{1}{648} \frac{F''(x)^3}{F'(x)^3}. \quad (3.5)$$

Proof. For $m = 1$, $c_1 = -a_1/F'$ and $a_1 = -b_1F''$. For $m = 2$,

$$c_2 = -\frac{a_2}{F'} + \frac{a_1 a_1'}{F'^2} - \frac{F'' a_1^2}{2F'^3}.$$

Insert $a_1 = -F''/18$ and $a_2 = 31F^{(4)}/16200$. $\square$

Remark 3.3. The forward expansion is linear in the derivative tower of F , but the inverse coefficients are necessarily nonlinear rational functions of that tower. The cubic term in (3.5) is not a higher Fourier correction; it is created solely by series reversion. This distinction is why simply applying the forward Richardson formula to a numerically inverted value does not reveal the correct leading constant.

3.3 Gaussian q -binomial Richardson filters for quantiles

For $s \geq 1$ and $0 \leq j \leq s-1$, define the Lagrange weight for evaluation at zero on the geometric nodes $1, q, \dots, q^{s-1}$ by

$$\lambda_{s,j} := \prod_{\substack{0 \leq \ell \leq s-1 \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell}, \quad q = \frac{1}{4}. \quad (3.6)$$

Factoring the products gives

$$\lambda_{s,j} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_j (q; q)_{s-1-j}} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_{s-1}} \begin{bmatrix} s-1 \\ j \end{bmatrix}_q. \quad (3.7)$$

In addition to the usual cancellations, the complete residual moments have a closed form.

Lemma 3.4 (All geometric moments of the Richardson filter). *For every integer $k \geq 1$,*

$$\sum_{j=0}^{s-1} \lambda_{s,j} q^{kj} = (-1)^{s-1} q^{\binom{s}{2}} \begin{bmatrix} k-1 \\ s-1 \end{bmatrix}_q, \quad (3.8)$$

where the Gaussian binomial is zero when $k < s$. Also $\sum_j \lambda_{s,j} = 1$.

Proof. The vanishing for $1 \leq k < s$ is Lagrange exactness. Substitute (3.7) in the left side, reverse the index, and apply the finite q -binomial theorem [7, 8] to the resulting alternating sum. This yields (3.8); at $k = s$ it reduces to $(-1)^{s-1} q^{\binom{s}{2}}$, the familiar first uncanceled moment. $\square$

Define the extrapolated quantile

$$\mathcal{G}_{n,s}(y) := \sum_{j=0}^{s-1} \lambda_{s,j} G_{n+j}(y), \quad G_m := P_m^{-1}. \quad (3.9)$$

Theorem 3.5 (q -binomial acceleration of finite-prefix quantiles). *Uniformly on every $[\eta, 1-\eta] \Subset (0, 1)$,*

$$\mathcal{G}_{n,s}(y) = G(y) + (-1)^{s-1} c_s(y) q^{\binom{s}{2}} q^{sn} + \mathcal{O}_{\eta,s}(q^{(s+1)n}). \quad (3.10)$$

More generally, if the full inverse germ is inserted, the coefficient of $c_k(y) q^{kn}$ for $k \geq s$ is the Gaussian-binomial moment in (3.8).

Proof. Insert (3.2) for each depth $n + j$ and interchange the finite sums. Formula (3.8) cancels every $k < s$ term and gives the displayed leading coefficient at $k = s$. $\square$

The first filters are

$$\mathcal{G}_{n,2} = \frac{4G_{n+1} - G_n}{3}, \quad (3.11)$$

$$\mathcal{G}_{n,3} = \frac{G_n - 20G_{n+1} + 64G_{n+2}}{45}, \quad (3.12)$$

$$\mathcal{G}_{n,4} = \frac{-G_n + 84G_{n+1} - 1344G_{n+2} + 4096G_{n+3}}{2835}. \quad (3.13)$$

They have orders 16^{-n} , 64^{-n} , and 256^{-n} respectively.

4 Exact inverse-dyadic local polynomials and algebraic quantile germs

4.1 The plateau mechanism

Put

$$\xi_r := -1 + 2^{-r} = x_r - 1, \quad a := 2^{-n}, \quad Q := a^2 = 4^{-n}. \quad (4.1)$$

The corpus proves the exact Thue–Morse derivative plateau

$$p_n^{(r)}(x) = 2^{\binom{r+1}{2}} \quad \text{for almost every } x \in [\xi_r - a, \xi_r + a], \quad n \geq r + 1. \quad (4.2)$$

Therefore $z \mapsto p_n(\xi_r + z)$ is an ordinary polynomial of degree at most r on $[-a, a]$. The next theorem identifies that polynomial exactly.

Let $M(t) = \mathbb{E}e^{tY}$ be the moment generating function of Y . Since $M(t) = \Phi(it)$, (2.10) gives

$$\frac{1}{M(t)} = \sum_{j=0}^{\infty} (-1)^j b_j t^{2j}. \quad (4.3)$$

Theorem 4.1 (Exact reciprocal-sinc local polynomial). *For $r \geq 1$, $n \geq r + 1$, and $|z| \leq 2^{-n}$,*

$$\boxed{P_n(x_r + z) = y_r + \mathcal{A}_r(z, Q)}, \quad (4.4)$$

where

$$\boxed{\mathcal{A}_r(z, Q) := J_r(z) + \sum_{j=1}^{\lfloor r/2 \rfloor} (-1)^j b_j Q^j J_r^{(2j)}(z)}. \quad (4.5)$$

Thus a formally infinite reciprocal-sinc operator becomes an exact finite operator on the terminating inverse-dyadic jet.

Proof. Set $q_n(z) := p_n(\xi_r + z)$ on $[-a, a]$. By (4.2), q_n is a polynomial of degree at most r . The tail factorization $Y = S_n + aY'$ with $Y' \stackrel{d}{=} Y$ gives

$$\text{up}(x) = \mathbb{E} p_n(x - aY). \quad (4.6)$$

Differentiating at $x = \xi_r$ shows, for $0 \leq k \leq r$,

$$\text{up}^{(k)}(\xi_r) = \mathbb{E} q_n^{(k)}(-aY). \quad (4.7)$$

The polynomial $M(-aD)q_n = M(aD)q_n$ has degree at most r ; its k th derivative at zero is the right side of (4.7). Hence, as polynomials,

$$M(aD)q_n(z) = \sum_{k=0}^r \frac{\text{up}^{(k)}(\xi_r)}{k!} z^k = y_r + J_r(z). \quad (4.8)$$

Apply the inverse operator (4.3). Derivatives of order greater than r vanish, so the series truncates exactly:

$$q_n(z) = \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j b_j a^{2j} D^{2j}(y_r + J_r(z)).$$

Finally use $P_n(x_r + z) = p_n(\xi_r + z)$. □

Remark 4.2 (Why this is stronger than the global asymptotic). The uniform expansion (2.16) has an arbitrarily high but nonzero remainder. At the plateau center, the exact polynomial argument bypasses that remainder completely. The identity is valid at finite n throughout a shrinking interval of radius 2^{-n} , not merely coefficientwise as $n \rightarrow \infty$.

4.2 The algebraic inverse branch

Theorem 4.3 (Exact algebraic quantile germ). *For each fixed $r \geq 1$, there is a unique real-analytic function $\Delta_r(Q)$ near $Q = 0$ satisfying*

$$\Delta_r(0) = 0, \quad \mathcal{A}_r(\Delta_r(Q), Q) = 0. \quad (4.9)$$

For all sufficiently large n ,

$$\boxed{G_n(y_r) = x_r + \Delta_r(4^{-n}).} \quad (4.10)$$

The function Δ_r is algebraic over $\mathbb{Q}(Q)$, and its Taylor coefficients are rational.

Proof. At $(z, Q) = (0, 0)$, $\mathcal{A}_r = 0$ and

$$\partial_z \mathcal{A}_r(0, 0) = J'_r(0) = F'(x_r) > 0.$$

The analytic implicit-function theorem gives a unique analytic germ Δ_r . Because (4.5) is a polynomial with rational coefficients, Δ_r is algebraic and its recursive Taylor coefficients are rational. Moreover $\Delta_r(Q) = \mathcal{O}(Q)$, so $|\Delta_r(4^{-n})| < 2^{-n}$ for all sufficiently large n . On that interval (4.4) applies, and its derivative remains positive near zero. The unique local solution of $P_n(x) = y_r$ is therefore exactly $x_r + \Delta_r(4^{-n})$. □

If

$$\Delta_r(Q) = \sum_{m \geq 1} \delta_{r,m} Q^m, \quad (4.11)$$

then the coefficients can be obtained without radicals:

$$\boxed{\delta_{r,m} = -\frac{1}{F'(x_r)} [Q^m] \mathcal{A}_r \left(\sum_{k=1}^{m-1} \delta_{r,k} Q^k, Q \right).} \quad (4.12)$$

This is a finite implicit version of (3.3). Expanding the powers of the truncated series recovers partial Bell polynomials.

4.3 The first four algebraic germs

The first jet polynomials are

$$J_1(z) = 2z, \quad (4.13)$$

$$J_2(z) = z + 4z^2, \quad (4.14)$$

$$J_3(z) = \frac{5}{36}z + 2z^2 + \frac{32}{3}z^3, \quad (4.15)$$

$$J_4(z) = \frac{1}{144}z + \frac{5}{18}z^2 + \frac{16}{3}z^3 + \frac{128}{3}z^4. \quad (4.16)$$

Consequently,

$$\mathcal{A}_1(z, Q) = 2z, \quad (4.17)$$

$$\mathcal{A}_2(z, Q) = z + 4z^2 - \frac{4}{9}Q, \quad (4.18)$$

$$\mathcal{A}_3(z, Q) = \frac{5}{36}z + 2z^2 + \frac{32}{3}z^3 - \frac{2}{9}Q - \frac{32}{9}Qz, \quad (4.19)$$

$$\begin{aligned} \mathcal{A}_4(z, Q) = & \frac{1}{144}z + \frac{5}{18}z^2 + \frac{16}{3}z^3 + \frac{128}{3}z^4 \\ & - \frac{5}{162}Q - \frac{16}{9}Qz - \frac{256}{9}Qz^2 + \frac{3968}{2025}Q^2. \end{aligned} \quad (4.20)$$

The corresponding outputs are

$$y_1 = \frac{1}{2}, \quad y_2 = \frac{5}{72}, \quad y_3 = \frac{1}{288}, \quad y_4 = \frac{143}{2073600}. \quad (4.21)$$

Thus $G_n(1/288)$ is, for large n , the small real root of the cubic (4.19), while $G_n(143/2073600)$ is governed by the quartic (4.20). Their first coefficients are

$$\Delta_3(Q) = \frac{8}{5}Q + \frac{512}{125}Q^2 - \frac{1245184}{3125}Q^3 + \mathcal{O}(Q^4), \quad (4.22)$$

$$\Delta_4(Q) = \frac{40}{9}Q + \frac{132608}{2025}Q^2 + \frac{126943232}{18225}Q^3 + \mathcal{O}(Q^4). \quad (4.23)$$

4.4 The quarter anchor: an exact square root

At $r = 2$, (4.18) is quadratic and can be solved without any asymptotic step.

Theorem 4.4 (Exact quarter-quantile formula). *For every $n \geq 3$,*

$$\boxed{G_n\left(\frac{5}{72}\right) = \frac{1}{4} + \frac{\sqrt{1 + \frac{64}{9}4^{-n}} - 1}{8}}. \quad (4.24)$$

Proof. For $r = 2$ and $n \geq 3$, the positive root

$$z_n = \frac{\sqrt{1 + (64/9)4^{-n}} - 1}{8}$$

satisfies $0 < z_n < 2^{-n}$. The exact local identity

$$P_n(1/4 + z) = \frac{5}{72} + z + 4z^2 - \frac{4}{9}4^{-n}$$

therefore applies at $z = z_n$, and the right side equals $5/72$. Positivity of the derivative $1 + 8z$ gives uniqueness on the local branch. $\square$

The singularity of the square root is at $Q = -9/64$, so its Taylor series converges for $|Q| < 9/64$. Let $C_m = \frac{1}{m+1}\binom{2m}{m}$ be the Catalan number. Then

$$\boxed{\Delta_2(Q) = \sum_{m=1}^{\infty} (-1)^{m-1} C_{m-1} \frac{2^{4m-2}}{9^m} Q^m}. \quad (4.25)$$

The first terms are

$$\Delta_2(Q) = \frac{4}{9}Q - \frac{64}{81}Q^2 + \frac{2048}{729}Q^3 - \frac{81920}{6561}Q^4 + \frac{3670016}{59049}Q^5 - \dots \quad (4.26)$$

This is a direct and previously absent bridge from the inverse Fabius problem to Catalan numbers.

4.5 Catalan–Gaussian-binomial Richardson constants

Apply the s -level filter (3.9) to the exact quarter-quantile sequence. Combining (4.25) and (3.8) gives the full convergent residual series

$$\mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} = (-1)^{s-1} q^{\binom{s}{2}} \sum_{m=s}^{\infty} \delta_{2,m} \begin{bmatrix} m-1 \\ s-1 \end{bmatrix}_q q^{mn}, \quad q = \frac{1}{4}. \quad (4.27)$$

In particular:

Corollary 4.5 (Closed leading constant at every Richardson order). *For every fixed $s \geq 1$,*

$$\mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} = \frac{C_{s-1} 2^{-s^2+5s-2}}{9^s} 4^{-sn} + \mathcal{O}(4^{-(s+1)n}). \quad (4.28)$$

The leading constant is positive for every s .

Proof. Use the $m = s$ term of (4.27), for which the Gaussian binomial equals one, and insert $\delta_{2,s} = (-1)^{s-1} C_{s-1} 2^{4s-2}/9^s$ and $q^{\binom{s}{2}} = 2^{-s(s-1)}$. $\square$

The first three error laws are therefore

$$G_n(5/72) - \frac{1}{4} \sim \frac{4}{9} 4^{-n}, \quad (4.29)$$

$$\mathcal{G}_{n,2}(5/72) - \frac{1}{4} \sim \frac{16}{81} 16^{-n}, \quad (4.30)$$

$$\mathcal{G}_{n,3}(5/72) - \frac{1}{4} \sim \frac{32}{729} 64^{-n}. \quad (4.31)$$

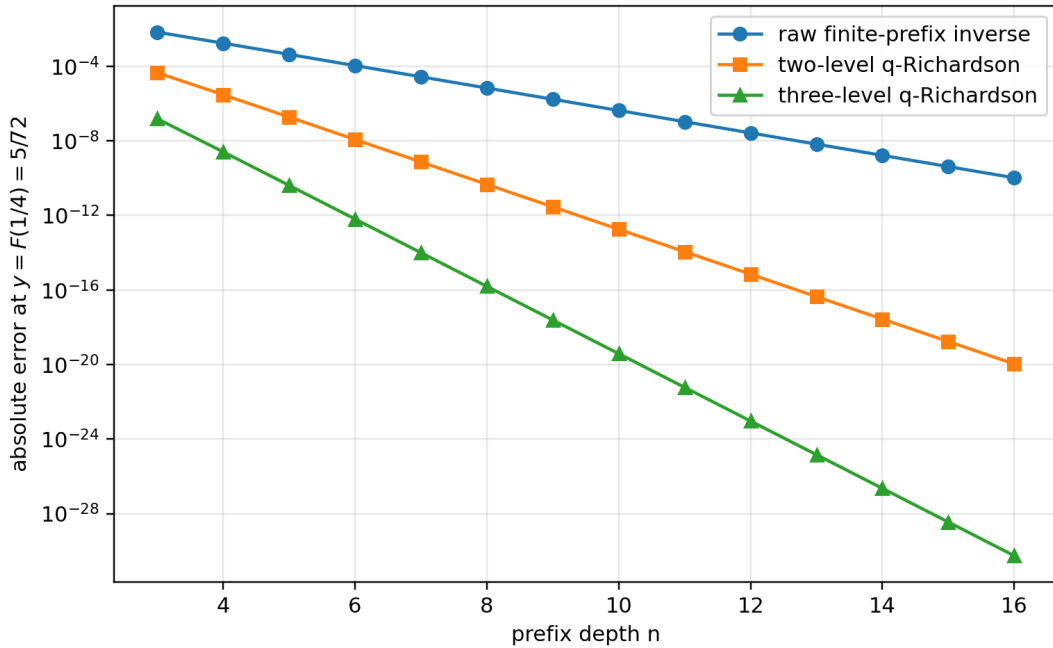


Figure 1: Raw and accelerated errors at the exact output $F(1/4) = 5/72$. The straight-line separation on the logarithmic scale reflects the exact geometric orders 4^{-n} , 16^{-n} , and 64^{-n} .

4.6 Exact sign and two-sided bounds for every accelerated quarter error

The preceding asymptotic constants can be strengthened to a finite- n interpolation statement. Put

$$f(Q) := \Delta_2(Q) = \frac{\sqrt{1 + (64/9)Q} - 1}{8}. \quad (4.32)$$

For fixed n and s , the quantity $\mathcal{G}_{n,s}(5/72) - 1/4$ is the value at $Q = 0$ of the polynomial of degree at most $s - 1$ interpolating f at the nodes $4^{-n}, 4^{-(n+1)}, \dots, 4^{-(n+s-1)}$.

Theorem 4.6 (Positive accelerated errors and sharp finite-depth bounds). *Let*

$$K_s := \frac{C_{s-1} 2^{-s^2+5s-2}}{9^s}. \quad (4.33)$$

For every $s \geq 1$ and $n \geq 3$,

$$0 < K_s 4^{-sn} \left(1 + \frac{64}{9} 4^{-n}\right)^{\frac{1}{2}-s} < \mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} < K_s 4^{-sn}. \quad (4.34)$$

In particular, the ratio of the exact error to its leading term is squeezed to one inside an explicitly shrinking enclosure.

Proof. Let $p_{n,s-1}$ be the Lagrange interpolant just described. The interpolation remainder at zero gives a point $\xi_{n,s} \in (0, 4^{-n})$ such that

$$-p_{n,s-1}(0) = \frac{f^{(s)}(\xi_{n,s})}{s!} \prod_{j=0}^{s-1} (-4^{-(n+j)}). \quad (4.35)$$

Now

$$f^{(s)}(Q) = \frac{1}{8} \left(\frac{1}{2}\right)^s \left(\frac{64}{9}\right)^s \left(1 + \frac{64}{9} Q\right)^{\frac{1}{2}-s}, \quad (4.36)$$

where the falling factorial has sign $(-1)^{s-1}$. The signs in (4.35) therefore imply $p_{n,s-1}(0) > 0$. Using

$$\frac{|(\frac{1}{2})^s|}{8s!} \left(\frac{64}{9}\right)^s 4^{-\binom{s}{2}} = K_s \quad (4.37)$$

produces the middle expression in (4.34) with $\xi_{n,s}$ in place of either endpoint. Since the exponent $1/2 - s$ is negative, evaluation at 0 and 4^{-n} gives the two strict bounds. $\square$

Remark 4.7. The positivity in [theorem 4.6](#) is not visible term-by-term in (4.27), whose tail alternates. It is a consequence of the complete monotonicity pattern of the derivatives of the square-root germ and the sign of the Lagrange remainder. This is one concrete place where exact interpolation gives more information than formal Richardson cancellation.

Proposition 4.8 (Large-order Catalan coefficients). *The Taylor coefficients of the quarter germ satisfy*

$$\delta_{2,m} \sim (-1)^{m-1} \frac{1}{16\sqrt{\pi}} \left(\frac{64}{9}\right)^m m^{-3/2}. \quad (4.38)$$

Thus the branch point $Q = -9/64$ controls both the convergence radius and the $m^{-3/2}$ algebraic prefactor.

Proof. Insert the classical Catalan asymptotic $C_{m-1} \sim 4^{m-1}/(\sqrt{\pi} m^{3/2})$ into (4.25). $\square$

4.7 A universal first coefficient at every inverse-dyadic anchor

The leading displacement of the algebraic germ admits a particularly compact expression in the dyadic moment numbers d_r from (2.19).

Proposition 4.9 (First algebraic-germ coefficient). *For every $r \geq 2$,*

$$\delta_{r,1} = \frac{1}{18} \frac{F''(2^{-r})}{F'(2^{-r})} = \frac{2^{r-1}(r-1)d_{r-2}}{9d_{r-1}} > 0. \quad (4.39)$$

For $r = 1$, $\Delta_1 \equiv 0$ and $G_n(1/2) = 1/2$ exactly.

Proof. The first equality is (3.4) evaluated at x_r . Formula (2.18) gives

$$\frac{F''(x_r)}{F'(x_r)} = 4 \frac{F(2^{2-r})}{F(2^{1-r})}.$$

Substitute (2.19) at indices $r - 2$ and $r - 1$ and simplify the powers of two. Positivity follows from $d_j > 0$. The case $r = 1$ follows from symmetry of every prefix density. $\square$

The values $4/9$, $8/5$, and $40/9$ in (4.26), (4.22), and (4.23) are therefore not isolated coincidences; they are the first members of a single exact rational sequence controlled by adjacent dyadic moments.

5 Periodic Lambert–Bell inversion at the endpoint

5.1 The forward input and its periodic data

We now invert the all-orders small- x expansion already established in the corpus. Put

$$L := \log 2, \quad B := 1 + \frac{L}{2}, \quad \beta := \frac{B}{L} = \frac{1}{L} + \frac{1}{2}. \quad (5.1)$$

For sufficiently small $x > 0$, the exact lower-Lambert phase is the large positive solution

$$x = \lambda 2^{-\lambda} = \lambda e^{-L\lambda}, \quad \lambda(x) = -\frac{1}{L} W_{-1}(-Lx)[9]. \quad (5.2)$$

The sharp forward expansion has, for every fixed N , the form

$$\log F(x) = -\frac{L}{2} \lambda^2 + B\lambda - \frac{1}{2} \log \lambda + C_{\sharp} + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}). \quad (5.3)$$

Here every A_j and Ψ is smooth, bounded, and one-periodic. With γ denoting Euler’s constant and γ_1 the first Stieltjes constant in the convention $\zeta(s) = \frac{1}{s-1} + \sum_{m \geq 0} (-1)^m \gamma_m (s-1)^m / m!$,

$$C_{\sharp} = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (5.4)$$

The zero-mean periodic fluctuation has Fourier coefficients

$$\Psi(u) = \sum_{k \neq 0} \widehat{\Psi}(k) e^{2\pi i k u}, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}. \quad (5.5)$$

The first logarithmic correction is

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (5.6)$$

The Gamma factor in (5.5) gives exponential Fourier decay, so all phase differentiations below are legitimate.

5.2 Two-scale inversion of the saddle phase

Let

$$T := \log \frac{1}{y}, \quad \rho := \sqrt{\frac{2T}{L}}, \quad \theta := \rho + \beta. \quad (5.7)$$

The leading inverse equivalent from the corpus is conveniently written

$$G_0(y) := \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}. \quad (5.8)$$

The following theorem computes what the corpus left implicit.

Theorem 5.1 (Phase inversion through two nontrivial orders). *As $y \downarrow 0$, the exact phase of $x = G(y)$ satisfies*

$$\lambda(G(y)) = \rho + \beta + \frac{d_1}{\rho} + \frac{d_2}{\rho^2} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^3}\right), \quad (5.9)$$

where all periodic functions below are evaluated at θ and

$$d_1 = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta) - \frac{1}{2} \log \rho}{L}, \quad (5.10)$$

$$d_2 = \frac{\Psi'(\theta)d_1 + A_1(\theta) - \frac{1}{2}\beta}{L}. \quad (5.11)$$

Consequently,

$$\log \frac{G(y)}{G_0(y)} = \frac{h_1}{\rho} + \frac{h_2}{\rho^2} + \mathcal{O}\left(\frac{(\log \rho)^3}{\rho^3}\right), \quad (5.12)$$

with

$$h_1 = \beta - Ld_1 = \frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta), \quad (5.13)$$

$$h_2 = d_1 - \frac{\beta^2}{2} - Ld_2 = (1 - \Psi'(\theta))d_1 - \frac{\beta^2}{2} - A_1(\theta) + \frac{\beta}{2}, \quad (5.14)$$

$$\kappa_0 := \beta - \frac{L}{2}\beta^2 - C_\# = \frac{1}{2L} - \frac{L}{8} - C_\#. \quad (5.15)$$

Equivalently, in multiplicative form,

$$\frac{G(y)}{G_0(y)} = 1 + \frac{h_1}{\rho} + \frac{h_2 + \frac{1}{2}h_1^2}{\rho^2} + \mathcal{O}\left(\frac{(\log \rho)^3}{\rho^3}\right). \quad (5.16)$$

Proof. Negate (5.3) and use $T = L\rho^2/2$. Write

$$\lambda = \rho + \beta + D, \quad D = \frac{d_1}{\rho} + \frac{d_2}{\rho^2} + \dots. \quad (5.17)$$

The quadratic part simplifies exactly:

$$\frac{L}{2}\lambda^2 - B\lambda = \frac{L}{2}\rho^2 + L\rho D - \frac{L}{2}\beta^2 + \frac{L}{2}D^2. \quad (5.18)$$

The constant term after cancellation of $L\rho^2/2$ is

$$Ld_1 - \frac{L}{2}\beta^2 + \frac{1}{2} \log \rho - C_\# - \Psi(\theta),$$

which gives (5.10). The coefficient of ρ^{-1} is

$$Ld_2 + \frac{\beta}{2} - \Psi'(\theta)d_1 - A_1(\theta),$$

which gives (5.11). The all-orders remainder in (5.3), together with $d_1 = \mathcal{O}(\log \rho)$, then yields the stated phase remainder by one Newton correction; the derivative of the phase equation is $L\rho + \mathcal{O}(1)$.

Finally,

$$\begin{aligned} \log G(y) &= \log \lambda - L\lambda \\ &= \log \rho - L\rho - L\beta + \log\left(1 + \frac{\beta}{\rho} + \frac{D}{\rho}\right) - LD. \end{aligned}$$

Expansion through ρ^{-2} gives (5.13)–(5.14); exponentiation gives (5.16). $\square$

The first consequence is the promised explicit correction to the leading equivalent.

Corollary 5.2 (First oscillatory relative correction). *As $y \downarrow 0$,*

$$\frac{G(y)}{G_0(y)} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\rho + \beta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right). \quad (5.19)$$

Remark 5.3 (Why the exact phase shift matters). Replacing $\rho + \beta$ by ρ , or replacing ρ by a cruder expression in $\log(1/y)$ inside Ψ , changes the phase by a fixed noninteger amount and therefore changes the correction at its full natural size $1/\rho$. The constant shift $\beta = 1/L + 1/2$ is not optional bookkeeping.

5.3 A complete periodic limit set

The oscillation in (5.19) cannot be absorbed into a constant second term.

Theorem 5.4 (Limit set of the normalized second term). *Define*

$$\mathcal{R}(y) := \rho \left(\frac{G(y)}{G_0(y)} - 1 \right) - \frac{1}{2} \log \rho. \quad (5.20)$$

Then the complete set of subsequential limits as $y \downarrow 0$ is

$$\text{range}_{y \downarrow 0} \mathcal{R}(y) = \left[\kappa_0 - \max_{u \in [0,1]} \Psi(u), \kappa_0 - \min_{u \in [0,1]} \Psi(u) \right]. \quad (5.21)$$

The interval is nondegenerate. Hence there is no constant c for which

$$\frac{G(y)}{G_0(y)} = 1 + \frac{\frac{1}{2} \log \rho + c}{\rho} + o(\rho^{-1}). \quad (5.22)$$

Proof. Corollary 5.2 gives

$$\mathcal{R}(y) = \kappa_0 - \Psi(\rho + \beta) + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho}\right). \quad (5.23)$$

As y ranges down to zero, ρ ranges continuously to $+\infty$. For any $u \in [0, 1)$, choose $\rho_m = m + u - \beta$ and $y_m = \exp(-L\rho_m^2/2)$. Then the phase in (5.23) is exactly u modulo one, proving that every value of $\kappa_0 - \Psi$ is a subsequential limit. Conversely, compactness of the phase circle shows that every sequence has a phase-convergent subsequence, so no other limit can occur.

The Fourier coefficient $\hat{\Psi}(1)$ in (5.5) is nonzero: Gamma has no zeros and the classical zero-free line $\Re s = 1$ gives $\zeta(1 - \chi_1) \neq 0$ [12]. Thus Ψ is nonconstant and the interval has positive length. Formula (5.22) would force a singleton limit set, a contradiction. $\square$

5.4 Quantile elasticity and inverse derivatives

The logarithmic derivative of the inverse is often more stable numerically than G' itself.

Theorem 5.5 (Two-term endpoint elasticity). *As $y \downarrow 0$,*

$$\boxed{\frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{\Psi'(\rho + \beta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right).} \quad (5.24)$$

Consequently,

$$G'(y) = \frac{G(y)}{y\rho} \left[1 + \frac{\Psi'(\rho + \beta) - 1}{L\rho} + \mathcal{O}\left(\frac{\log \rho}{\rho^2}\right) \right]. \quad (5.25)$$

Proof. Regard both $T = -\log y$ and $\log x$ as functions of the exact phase λ . Differentiating (5.3) gives

$$\frac{yG'(y)}{G(y)} = -\frac{d \log x / d\lambda}{dT / d\lambda} = \frac{L - \lambda^{-1}}{L\lambda - B + \frac{1}{2\lambda} - \Psi'(\lambda) + \mathcal{O}(\lambda^{-2})}. \quad (5.26)$$

Insert $\lambda = \rho + \beta + \mathcal{O}((\log \rho)/\rho)$ and expand. The terms containing β cancel from the denominator because $B = L\beta$, leaving (5.24). $\square$

By reflection, every formula in this section has a right-endpoint counterpart: replace y by $1 - y$ and $G(y)$ by $1 - G(y)$. In particular, the same periodic function and the same phase shift govern the two ends.

5.5 An all-orders Lambert–Bell reversion algorithm

The preceding calculation extends mechanically to every order, but ordinary power-series notation hides the two-scale nature of the coefficients. Treat $z = 1/\rho$ as the small variable while holding the fast phase $\theta = \rho + \beta$ and the slow logarithm $\log \rho$ as coefficient variables. Let

$$D(z) = \sum_{m \geq 1} d_m z^m, \quad \lambda = \rho + \beta + D(z). \quad (5.27)$$

Define

$$\begin{aligned} \mathcal{R}(z, D) := & -\frac{L}{2}\beta^2 + \frac{L}{2}D^2 + \frac{1}{2}\log \rho + \frac{1}{2}\log(1 + \beta z + zD) - C_{\sharp} - \Psi(\theta + D) \\ & - \sum_{j \geq 1} A_j(\theta + D)z^j(1 + \beta z + zD)^{-j}. \end{aligned} \quad (5.28)$$

The phase equation is formally

$$0 = \frac{L}{z}D + \mathcal{R}(z, D). \quad (5.29)$$

Theorem 5.6 (Triangular all-orders endpoint recursion). *Let $D_m(z) = \sum_{j=1}^m d_j z^j$ and set $D_0 = 0$. The unique formal solution of (5.29) is generated by*

$$\boxed{d_{m+1} = -\frac{1}{L}[z^m]\mathcal{R}(z, D_m), \quad m \geq 0.} \quad (5.30)$$

Each d_m is a polynomial in $\log \rho$ with coefficients in the differential algebra generated by the periodic functions $\Psi, A_1, \dots, A_{m-1}$.

Write

$$\log \frac{G(y)}{G_0(y)} = \log(1 + \beta z + zD) - LD = \sum_{m \geq 1} h_m z^m. \quad (5.31)$$

Then the multiplicative coefficients g_m in

$$\frac{G(y)}{G_0(y)} \sim 1 + \sum_{m \geq 1} g_m \rho^{-m} \quad (5.32)$$

are the complete Bell polynomials

$$g_m = \frac{1}{m!} \mathcal{B}_m(1!h_1, 2!h_2, \dots, m!h_m). \quad (5.33)$$

For every fixed truncation order, the resulting expansion is rigorous with a remainder uniform in the phase.

Proof. The coefficient of d_{m+1} in $[z^m](LD/z)$ is exactly L , while $[z^m]\mathcal{R}$ depends only on $d_1, \dots, d_m$. This gives the triangular recursion. Taylor expansion of each periodic coefficient at θ shows inductively that the coefficient algebra is closed under the required operations. Formula (5.33) is the standard exponential formula applied to (5.31). The uniform all-orders remainder in (5.3), together with the fact that the derivative of the phase equation is asymptotic to $L\rho$, upgrades the formal induction to a Poincaré expansion at any fixed order. $\square$

The endpoint inverse has a mixed asymptotic architecture:

lower Lambert W + polynomials in $\log \rho$ + periodic Gamma–zeta jets + Bell reversion.

None of these four layers can be removed without losing a term at some finite order.

6 Symbolic, exact, and high-precision verification

6.1 Exact rational checks of the local-polynomial theorem

The construction of $\mathcal{A}_r$ was checked independently in exact rational arithmetic by the following pipeline:

1. generate $d_0, \dots, d_r$ from the recurrence

$$d_m = \frac{\sum_{k=0}^{m-1} \binom{m+1}{k} d_k}{(m+1)(2^m - 1)}; \quad (6.1)$$

2. form the dyadic jet J_r from (2.20);
3. generate b_j from (2.14);
4. apply the finite operator in (4.5);
5. substitute the resulting root series into $\mathcal{A}_r$ and verify vanishing coefficient-by-coefficient.

For $r \leq 12$, all coefficients through Q^{12} vanish exactly after substitution. For $r = 2$, direct substitution of the radical in (4.24) reduces the identity to zero without series expansion.

6.2 Quarter-anchor convergence and exact enclosures

The table illustrates a useful distinction. Richardson acceleration does not merely make the error small: the exact square-root germ supplies a certified sign and a certified finite- n relative enclosure. At $n = 8$, three levels already reduce the absolute error below 1.6×10^{-16} .

n	raw error	ratio	$s = 2$ error	ratio	$s = 3$ error	ratio
3	$6.76157 \cdot 10^{-3}$	0.973666	$4.51028 \cdot 10^{-5}$	0.935251	$1.53350 \cdot 10^{-7}$	0.915801
4	$1.72422 \cdot 10^{-3}$	0.993150	$2.96269 \cdot 10^{-6}$	0.982949	$2.55798 \cdot 10^{-9}$	0.977675
5	$4.33277 \cdot 10^{-4}$	0.998270	$1.87566 \cdot 10^{-7}$	0.995679	$4.06494 \cdot 10^{-11}$	0.994333
6	$1.08460 \cdot 10^{-4}$	0.999566	$1.17610 \cdot 10^{-8}$	0.998916	$6.37859 \cdot 10^{-13}$	0.998578
7	$2.71238 \cdot 10^{-5}$	0.999892	$7.35660 \cdot 10^{-10}$	0.999729	$9.97719 \cdot 10^{-15}$	0.999644
8	$6.78150 \cdot 10^{-6}$	0.999973	$4.59881 \cdot 10^{-11}$	0.999932	$1.55935 \cdot 10^{-16}$	0.999911

Table 2: Exact quarter-quantile errors. Each ratio divides by the corresponding leading term in (4.29)–(4.31). The values lie inside the analytic enclosure (4.34).

6.3 Endpoint inversion on exact dyadic test values

For $y_n = F(2^{-n})$, the true inverse is exactly $G(y_n) = 2^{-n}$. The dyadic moment recurrence therefore provides endpoint test values without numerical root finding. Define

$$G^{[0]}(y) := G_0(y), \quad (6.2)$$

$$G^{[1]}(y) := G_0(y) \exp(h_1/\rho), \quad (6.3)$$

$$G^{[2]}(y) := G_0(y) \exp(h_1/\rho + h_2/\rho^2). \quad (6.4)$$

The periodic function and its first two derivatives were evaluated from (5.5) with nine positive Fourier modes at 100 decimal digits.

n	ρ	relative error of $G^{[0]}$	of $G^{[1]}$	of $G^{[2]}$
5	6.43984	$-3.82510 \cdot 10^{-1}$	$7.89489 \cdot 10^{-2}$	$-2.16957 \cdot 10^{-2}$
10	12.08040	$-2.56964 \cdot 10^{-1}$	$2.68825 \cdot 10^{-2}$	$-4.39948 \cdot 10^{-3}$
20	22.82438	$-1.61765 \cdot 10^{-1}$	$8.76101 \cdot 10^{-3}$	$-8.23920 \cdot 10^{-4}$
40	43.65176	$-9.65552 \cdot 10^{-2}$	$2.71602 \cdot 10^{-3}$	$-1.43280 \cdot 10^{-4}$
80	84.54102	$-5.53439 \cdot 10^{-2}$	$8.04968 \cdot 10^{-4}$	$-2.34032 \cdot 10^{-5}$
100	104.83719	$-4.59578 \cdot 10^{-2}$	$5.39809 \cdot 10^{-4}$	$-1.29189 \cdot 10^{-5}$

Table 3: Relative errors at the exact points $y_n = F(2^{-n})$. The first and second inverse corrections successively remove the dominant endpoint error.

Numerically,

$$\begin{aligned} L &= 0.6931471805599453094 \dots, \\ \beta &= 1.9426950408889634074 \dots, \\ C_{\sharp} &= -2.0279838986388594240 \dots, \\ \kappa_0 &= 2.6626880215133479640 \dots \end{aligned} \quad (6.5)$$

The first Fourier coefficient has

$$|\widehat{\Psi}(1)| = 1.0627109779221651 \times 10^{-6}, \quad \arg \widehat{\Psi}(1) = -0.7795055208985225 \dots, \quad (6.6)$$

and the second mode is smaller by a factor about 6.30×10^{-7} . Thus Ψ is very close to a single sinusoid, although its small amplitude does not make it asymptotically dispensable.

The numerical range in figure 3 turns the abstract interval (5.21) into an interval of width approximately 4.25085×10^{-6} centered extremely close to κ_0 . The width is tiny, but it is fixed; multiplying the relative error by ρ reveals it rather than erasing it.

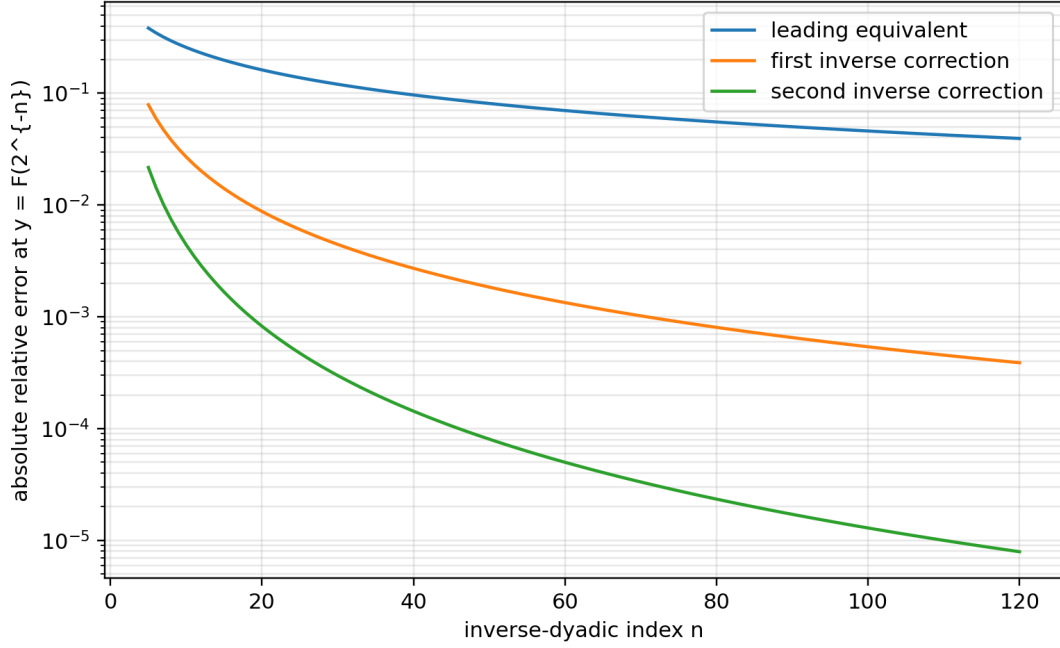


Figure 2: Absolute relative errors of the leading, first-corrected, and second-corrected endpoint inverse approximations on exact inverse-dyadic test values.

6.4 Independent checks and failure modes

Three independent representations were used wherever possible:

1. exact rational recurrences for d_r , y_r , J_r , and the coefficients of $\mathcal{A}_r$;
2. symbolic series reversion, including direct residual checks $\mathcal{A}_r(\Delta_r(Q), Q) = \mathcal{O}(Q^{M+1})$;
3. high-precision evaluation of the Gamma-zeta Fourier series and exact dyadic inverse test points.

The principal numerical failure mode is phase corruption: replacing the exact $\theta = \rho + \beta$ by a rounded or asymptotically simplified phase can dominate the very small periodic signal. The principal algebraic failure mode is confusing the forward coefficient $-b_1 F''$ with the inverse coefficient $b_1 F''/F'$; the missing Jacobian changes every Richardson constant.

7 Algorithms and a Lean formalization roadmap

7.1 Exact construction of an inverse-dyadic germ

The following pseudocode is entirely rational up to the final algebraic root or formal series operation.

Listing 1: Exact construction of the local polynomial and inverse germ.

```
input: anchor depth r, requested series order M

# Dyadic moment numbers
D[0] = 1
for m = 1..r:
    D[m] = sum(binomial(m+1,k)*D[k], k=0..m-1)
           / ((m+1)*(2^m-1))

# Reciprocal-sinc coefficients
```

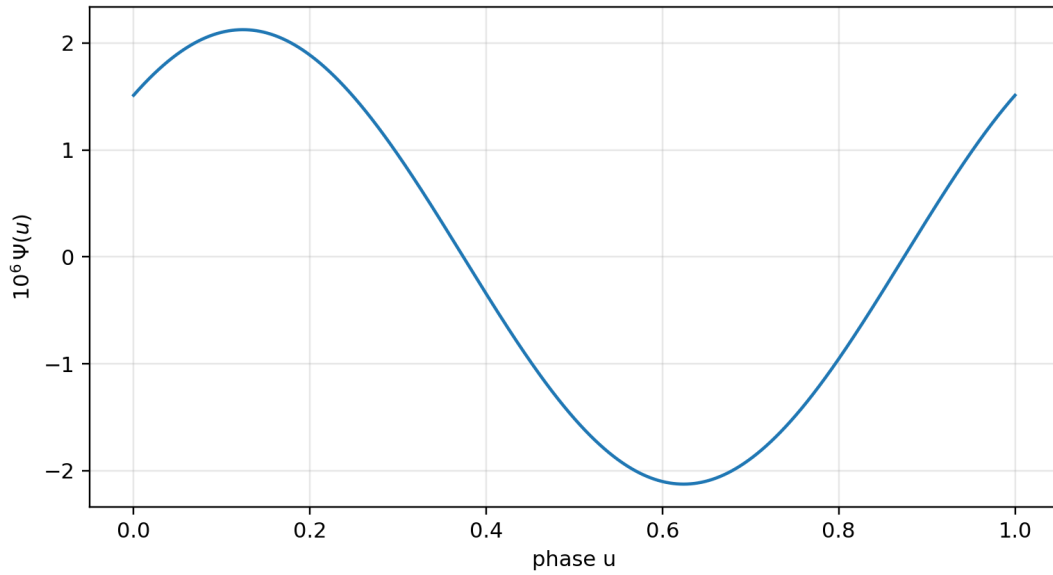


Figure 3: The periodic Gamma–zeta fluctuation. The vertical scale is multiplied by 10^6 . A grid evaluation gives approximately $\min \Psi = -2.1254264 \times 10^{-6}$ and $\max \Psi = 2.1254241 \times 10^{-6}$.

```

b[0] = 1
for j = 1..floor(r/2):
    gamma[j] = zeta(2*j)/(j*pi^(2*j)*(4^j-1)) # rational after simplification
    b[j] = sum(k*gamma[k]*b[j-k], k=1..j)/j

# Terminating Fabius jet
J(z) = sum(
    2^(binomial(k+1,2)-binomial(r-k,2))
    * D[r-k] * z^k / (k!*(r-k)!),
    k=1..r)

# Exact finite-prefix local polynomial
A(z,Q) = J(z)
for j = 1..floor(r/2):
    A(z,Q) += (-1)^j * b[j] * Q^j * derivative(J,z,2*j)

# Formal inverse branch A(Delta(Q),Q)=0
Delta(Q) = 0
for m = 1..M:
    residual = coefficient(A(Delta(Q),Q), Q, m)
    delta[m] = -residual / derivative(J,z,1).subs(z=0)
    Delta(Q) += delta[m]*Q^m

return A(z,Q), Delta(Q)

```

For $r = 2$, a radical solver should be used instead of formal reversion. For larger r , formal reversion is usually preferable to radicals: it preserves rationality, scales to high order, and feeds directly into the Gaussian-binomial residual formula.

7.2 Endpoint coefficient engine

The all-orders endpoint recursion can be implemented in a differential polynomial ring whose atoms are $\log \rho$, $\Psi^{(k)}(\theta)$, and $A_j^{(k)}(\theta)$.

Listing 2: Lambert–Bell inverse coefficient engine.

```

input: desired inverse order N; forward periodic coefficients A[1..N-1]
z = formal variable
D = 0
for m = 0..N-1:
  R = -L*beta^2/2 + L*D^2/2 + log(rho)/2
    + log(1 + beta*z + z*D)/2 - Csharp
    - Taylor_phase(Psi, theta, D)
    - sum(
      Taylor_phase(A[j], theta, D)
      * z^j * (1+beta*z+z*D)^(-j),
      j=1..N-1)
  d[m+1] = -coefficient(R, z, m)/L
  D += d[m+1]*z^(m+1)

H = truncate(log(1+beta*z+z*D)-L*D, z, N)
for m = 1..N:
  h[m] = coefficient(H, z, m)
  g[m] = CompleteBell(m, 1!*h[1], ..., m!*h[m]) / m!
return d, h, g

```

The supplied reproducibility script evaluates the first two orders directly rather than building a general symbolic differential ring, but its formulas are exact transcriptions of [theorem 5.1](#).

7.3 Formalization modules

The results separate naturally into five Lean-sized packages.

Formalization obligation 7.1 (Polynomial smoothing and inverse moment operator). Formalize that, for a polynomial q of degree at most r ,

$$M(aD)q = J \implies q = \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j b_j a^{2j} D^{2j} J.$$

This requires only finite sums after the degree bound is known; no convergence theorem for formal differential operators is needed.

Formalization obligation 7.2 (Plateau-to-polynomial localization). Connect the existing almost-everywhere derivative plateau for $p_n^{(r)}$ to the statement that p_n agrees with a degree- r polynomial on the closed interval $[\xi_r - 2^{-n}, \xi_r + 2^{-n}]$. The clean route is repeated absolute continuity of the spline pieces.

Formalization obligation 7.3 (Algebraic inverse germ). Instantiate a formal implicit-function theorem over $\mathbb{Q}[[Q]]$ for $\mathcal{A}_r(z, Q)$ with invertible linear coefficient $F'(x_r)$. Prove separately that the formal series is represented by the analytic local inverse and enters the plateau interval for all large n .

Formalization obligation 7.4 (Analytic use of formalized geometric moments). The finite all-filter moment identity, its quarter-base specialization, and the residual sign kernel are now formalized in `GeometricQBinomialLagrange.lean` and `GeometricLagrangeQMoments.lean`. It remains to instantiate the inverse analytic series, justify the required infinite-series interchange, and transfer the finite sign identity to the full quarter-error. That last step still needs the analytic Lagrange remainder formula in the report’s domain.

Formalization obligation 7.5 (Two-scale endpoint reversion). Build a small asymptotic algebra in the variables ρ^{-1} , $\log \rho$, and a phase on $\mathbb{R}/\mathbb{Z}$. The first target should be (5.19); it needs only the forward expansion through A_1 and elementary Taylor bounds. The all-orders recursion can then be layered on top of the already formalized forward saddle expansion.

The first four obligations are largely algebraic and should be substantially easier than the existing saddle analysis. The fifth is analytically deeper, but the forward uniform remainder estimates already present in the development remove its most difficult input.

8 Further deductions and frontier problems

8.1 Discriminants and convergence radii of the algebraic germs

For fixed r , the radius of convergence of $\Delta_r(Q)$ is bounded by the nearest complex point where the local branch ceases to be simple. Algebraically, candidate singularities are obtained by eliminating z from

$$\mathcal{A}_r(z, Q) = 0, \quad \partial_z \mathcal{A}_r(z, Q) = 0. \quad (8.1)$$

For $r = 2$ this gives $Q = -9/64$. For $r = 3$ and $r = 4$, the resulting discriminants are explicit but already nontrivial. Determining their nearest roots would give exact large-order laws for $\delta_{r,m}$ and sharp convergence domains for the inverse-prefix expansions.

8.2 Validity thresholds

The proof of [theorem 4.3](#) yields an unspecified threshold $n_0(r)$ from the condition $|\Delta_r(4^{-n})| < 2^{-n}$. Computation strongly suggests that the natural plateau threshold

$$n \geq r + 1 \quad (8.2)$$

may already suffice for every r . A proof would require a uniform root bound for the small branch of $\mathcal{A}_r$ in terms of the positive dyadic moment ratios. The exact positivity of [\(4.39\)](#) is encouraging but does not by itself control higher terms.

Conjecture 8.1 (Natural threshold for inverse-dyadic germs). *For every $r \geq 1$ and every $n \geq r + 1$, the small real root $\Delta_r(4^{-n})$ lies in $[-2^{-n}, 2^{-n}]$ and*

$$G_n(F(2^{-r})) = 2^{-r} + \Delta_r(4^{-n}).$$

8.3 Stieltjes and Bernstein structure

The quarter germ is a complete Bernstein function of $Q > 0$. This explains the alternating Catalan coefficients, positivity of every extrapolated error, and the interpolation bounds in [theorem 4.6](#). It is natural to ask which of the higher algebraic germs Δ_r belong to a Stieltjes, Bernstein, or Pick class after an elementary normalization. A positive answer would turn sign patterns that are currently empirical into global theorems and might supply Padé bounds stronger than Richardson extrapolation.

8.4 Uniform transition between finite-prefix and endpoint regimes

The expansion [\(3.2\)](#) is uniform only when the output remains in a compact subinterval of $(0, 1)$, while [section 5](#) treats the exact inverse G as $y \downarrow 0$. A two-parameter frontier problem is to understand

$$G_n(y_n) - G(y_n) \quad \text{when} \quad y_n \downarrow 0 \quad (8.3)$$

with n simultaneously large. The crossover should occur when the omitted tail scale 2^{-n} is comparable with the endpoint saddle width. The exact local polynomial theorem supplies anchor points inside this transition, and the Lambert phase supplies its natural continuous coordinate.

8.5 Optimal phase-aware endpoint approximants

The first periodic mode of Ψ dominates all later modes by many orders of magnitude. One may therefore construct inexpensive approximants using only $\widehat{\Psi}(\pm 1)$ while retaining a rigorous tail bound from Stirling’s formula for Gamma and standard vertical-line estimates for zeta. A useful computational target is a certified endpoint inverse routine whose cost is essentially that of evaluating W_{-1} plus one complex exponential, with a user-prescribed relative error.

8.6 Inverse functional equations

The differential identity $F'(x) = 2F(2x)$ on the first half of the interval yields

$$G'(y) = \frac{1}{2F(2G(y))} \quad (0 < y < 1/2). \quad (8.4)$$

The right side can itself be expressed through G^{-1} , but not locally in $G(y)$ alone. A promising direction is to regard the graph of G as an invariant curve for a nonlinear renormalization operator. The endpoint expansion in this report gives a formal stable manifold for that operator, while the inverse-dyadic algebraic germs give exact finite-scale sections.

8.7 Publication-priority boundary

The corpus audit and targeted literature screening found extensive work on arithmetic values, Fourier products, and dyadic evaluation [4, 5], together with the repository’s leading inverse asymptotic, but no occurrence of the exact local polynomials (4.4), the Catalan quantile formula (4.24), or the explicit oscillatory inverse correction (5.19). This supports the claim of novelty relative to the repository and motivates external dissemination. It does not replace a full bibliographic priority search, especially across literature using different normalizations of the Fabius or up-function.

9 Conclusion

The Fabius–Rvachev system contains two complementary kinds of self-similarity. The first is finite and arithmetic: Thue–Morse inclusion–exclusion, dyadic derivative plateaux, rational reciprocal-sinc coefficients, and Gaussian q -binomial interpolation. The second is asymptotic and spectral: lower-Lambert inversion, Gamma–zeta periodicity, and Bell-polynomial series reversion. Passing from the function to its inverse makes these structures interact in ways not visible in the forward theory.

At finite scale, a terminating derivative jet and an exact plateau turn a formal smoothing inverse into a finite polynomial identity. This gives algebraic quantile germs at every inverse-dyadic anchor and, at the quarter anchor, an exact square root with Catalan coefficients. Lagrange interpolation on geometric nodes then produces Gaussian-binomial Richardson laws with closed constants, exact signs, and finite-depth enclosures.

At the endpoint, inverting one additional order beyond the known equivalent reveals the periodic fluctuation that the leading term suppresses. Its phase shift is exact, its subsequential limit set is a nondegenerate interval, and its higher coefficients are generated by a triangular Lambert–Bell recursion. The result is a coherent inverse calculus spanning exact arithmetic, asymptotic analysis, special functions, and extrapolation theory.

A Complete recursive LaTeX corpus inventory

The following ledger records the 57 inspected sources under `Analysis/FabiusFunction/docs`. Historical variants were retained for novelty screening even when superseded, because they sometimes contain derivations or conjectures not copied into the living document.

A.1 Living documents and embedded papers: 25 files

No.	Repository-relative path
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	non-formalized-research-frontiers/Non_Formalized_Research_Frontiers.tex
6	non-formalized-research-frontiers/drafts/Fabius_Thue_Morse_Formula_Atlas/Fabius_Thue_Morse_Formula_Atlas.tex
7	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
16	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
18	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
19	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
20	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
21	papers/arXiv-1502.06738v2/Lacunary_products.tex
22	papers/arXiv-1702.05442v1/09-Function.tex
23	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24	papers/arXiv-2211.13570v1/document.tex
25	papers/ubiq15/ubiq15-corrected.tex

A.2 Frozen archive: 32 files

No.	Path below archive/
1	early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
2	early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-folds ummationoverthesignedThue-Morsesequence.tex
3	standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Ele mentarity_of_the_Fabius_Function.tex
4	standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymp totics.tex
5	standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inver se-dyadic-closed-form.tex
6	primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function _and_Rvachev_Up.tex
7	primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Functi on_and_Rvachev_Up.tex
8	primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Functi on_and_Rvachev_Up.tex
9	primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.t ex
10	primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
11	primary-exposition/Fabius_function_and_up_function/Fabius_functio n_and_up_function.tex
12	primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function _Missing_Parts.tex
13	primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Functi on_Missing_Parts.tex
14	q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions. tex
15	q-calculus/fabius-q-special-functions/fabius-q-special-functions .tex
16	lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.t ex
17	lean-walkthrough/fabius_lean_walkthrough-2/fabius_lean_walkthrough .tex
18	glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathe matical_Glossary.tex
19	glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Ma thematical_Glossary.tex
20	research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_A symptotic_Bridge.tex
21	research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Represen tations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
22	research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dy adic_Formulae_to_Asymptotics.tex
23	research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Con nections.tex
24	research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_ Morse_Convergence_Rate.tex

No.	Path below archive/—continued
25	research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
26	research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
27	research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
28	research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
29	research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
30	research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
31	research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
32	research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Compact coefficient reference

B.1 Reciprocal-sinc coefficients

With γ_j from (2.11),

$$b_0 = 1, \quad b_m = \frac{1}{m} \sum_{j=1}^m j \gamma_j b_{m-j}. \quad (\text{B.1})$$

The first values are repeated here for computation:

m	0	1	2	3	4
b_m	1	$\frac{1}{18}$	$\frac{31}{16200}$	$\frac{2347}{42865200}$	$\frac{1904369}{1311675120000}$

(B.2)

B.2 Inverse-transfer coefficients

For a general forward germ $P_Q(x) = F(x) + \sum_{j \geq 1} a_j(x) Q^j$ and inverse $G_Q(y) = G(y) + \sum_{j \geq 1} c_j(y) Q^j$, evaluated at $x = G(y)$,

$$c_1 = -\frac{a_1}{F'}, \quad (\text{B.3})$$

$$c_2 = -\frac{a_2 + a'_1 c_1 + \frac{1}{2} F'' c_1^2}{F'}, \quad (\text{B.4})$$

$$c_3 = -\frac{1}{F'} \left(a_3 + a'_2 c_1 + a'_1 c_2 + \frac{1}{2} a''_1 c_1^2 + F'' c_1 c_2 + \frac{1}{6} F''' c_1^3 \right). \quad (\text{B.5})$$

These are the first rows of the Bell recursion (3.3).

B.3 Endpoint reference

For $T = \log(1/y)$,

$$\rho = \sqrt{\frac{2T}{L}}, \quad \theta = \rho + \frac{1}{L} + \frac{1}{2}, \quad G_0 = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}. \quad (\text{B.6})$$

The first two log-corrections are

$$h_1 = \frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta), \quad (\text{B.7})$$

$$h_2 = (1 - \Psi'(\theta)) \left[\frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta) - \frac{1}{2} \log \rho}{L} \right] - \frac{\beta^2}{2} - A_1(\theta) + \frac{\beta}{2}. \quad (\text{B.8})$$

Then $G \approx G_0 \exp(h_1/\rho + h_2/\rho^2)$.

C Reproducibility files

The archive accompanying this report contains:

- the present LaTeX source and compiled PDF;
- the three plotted PNG figures;
- `generate_data.py`, using exact Python fractions, `mpmath`, and `matplotlib`;
- CSV tables for the exact quarter errors and high-precision endpoint tests;
- a text file recording the numerical constants and sampled periodic extrema.

The Python script makes no network requests. Re-running it regenerates every numerical value and figure used in [section 6](#).

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